Project Design Phase
Solution Architecture

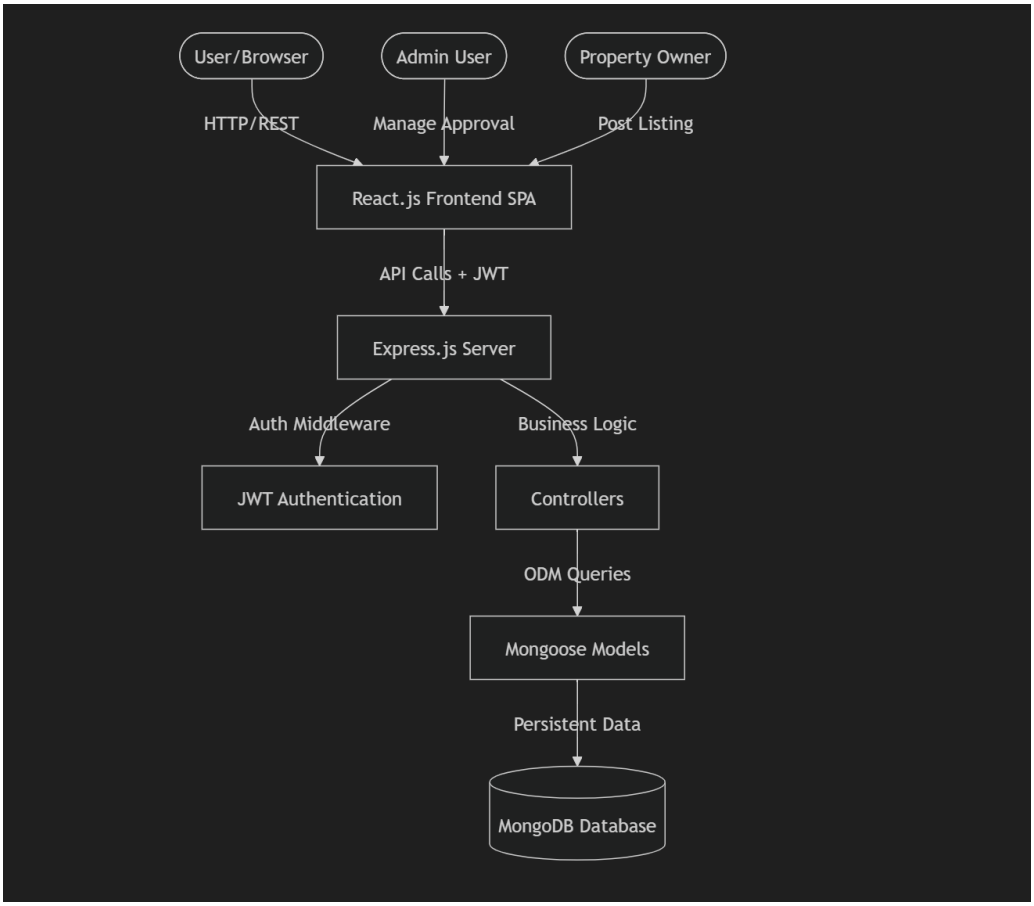
Date	15 February 2025
Team ID	699d5fb73074df695bbbb046
Project Name	HouseRent Management
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process — with many sub-processes — that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

HouseRent Management System Architecture Diagram:



Architecture Overview:

1. **Frontend: A responsive React single-page application built with modern UI components.**
2. **Backend: An Express.js REST API handling authentication, CRUD operations, and image processing via Multer.**
3. **Database: A flexible NoSQL MongoDB instance optimized for complex property and booking relationships.**
4. **Security: Stateless JWT-based sessions and bcrypt-hashed passwords to protect user identity and data.**